Project Management Plan

for

THEIA: Indoor Navigation App for the Blind and Visually Impaired

Version 1.0 draft 2

Prepared by Ubuntu Bros.

Reviewed by

Cpt S 484, Washington State University, Fall 2026

10/14/2026

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Revision History

Name	Date	Reason for Changes	Version
All	9/19/2026	initial draft	1.0 draft 1
Travis	10/14/2026	Revised Draft	1.0 draft 2

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1. Overview

This plan describes how Ubuntu Bros will carry out Phase I of the Cpt S 484 team project. In Phase I, the team will study the informal Preliminary Definition of THEIA, a smartphone app designed to help aid blind and visually impaired (BVI) people navigate indoors. Using the information gathered from the Preliminary Definition, the team will the produce a WRS document, AS-IS/TO-BE scenarios, a recorded presentation, a mock-up prototype, user manual, and an AI Usage and Teamwork report. The schedule is fixed to future course due dates and there is no budget; therefore the plan will be to divide the work fairly, keeping the work load coherent and streamlined to be completed at least a day before the due date to give the team time to edit and reread. The full planned schedule is available at Section 3.4.

1.1. Project Purpose, Objectives, and Success Criteria

Purpose. The purpose of the project is to practice software requirements engineering on a realistic problem at scale, while avoiding the dangers of working on a real project. This creates valuable experience through a simulated and controlled environment, allowing the team to learn the software requirements processes before working on real world projects.

Scope. This plan covers Phase I only. Phase I introduces requirements models, requirements specification, and a mock-up only prototype. It does not include a working implementation, design documents, or any other requirements an average project would require such as testing.

Objectives. The team has set the following measurable objectives for Phase I:

- **OBJ-1:** Identify and document issues in all three sections of Preliminary Definition and give a resolution and design rationale for every issue.
- **OBJ-2:** Produce an Improved Understanding in the WRS document that breaks down the document into Goals, Objectives, and Requirement specifications; each should have a flow of Requirement Specification traces to at least one Objective, every Objective to a Goal, and every Goal to a Problem.
- **OBJ-3:** Present four AS-IS/TO-BE scenario pairs and an analysis of the Top Priority TO-BE scenario in a recorded presentation.
- **OBJ-4:** Build a mock-up prototype that is composed of a series of UI sketches, rather than a full-fledged prototype, and a user manual that shows users common TO-BE scenarios
- **OBJ-5:** Submit a preliminary Project Plan for feedback, update with feedback during Phase I, and submit a final version at the end of Phase I.
- **OBJ-6:** Submit an AI Usage and Teamwork Report with evidence of any Gen AI use and a description of how the output has been reviewed

Success Criteria. The project will be considered successful if all six deliverables are submitted on or before their due dates; every document section has been reviewed by at least one team member other than its author; terminology and identifiers are consistent across all documents; instructor feedback on the preliminary Project Plan is incorporated into the final version; task distribution is reflected in each member's contributions.

Relationships to other work. The WRS document, user manual, and Project Plan will be further refined and finished in Phase II.

- **Shared resources**: team git repository, course templates, and samples on Canvas
- **Feature dependencies:**
    1. The Improved Understanding depends on the Issues and their design rationale.

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2. The TO-BE scenarios depend on the Requirements Specifications.
3. The prototype and user manual depend on the TO-BE scenarios.

- **Schedule Dependencies:** instructor feedback on preliminary plan is needed before finalized plan and continuing of rest of project

1.2. Project Deliverables

Deliverable	Delivery Date	Delivery Method	Comments
Preliminary Project Plan	9/20/2026	Canvas Submission/GitHub Repository	
WRS Document: Issues (III.2.1)	10/16/2026	Canvas Submission/GitHub Repository	Deadline given 2 days before actual Canvas deadline to give time for editing and revisions
WRS Document: Improved Understanding (III.2.2)	10/16/2026	Canvas Submission/GitHub Repository	Deadline given 2 days before actual Canvas deadline to give time for editing and revisions
AS-IS/TO-BE Slides and Presentation	10/16/2026	Canvas Submission/GitHub Repository	Deadline given 2 days before actual Canvas deadline to give time for editing and revisions
Project Prototype/User Manual	10/16/2026	Canvas Submission/GitHub Repository	Deadline given 2 days before actual Canvas deadline to give time for editing and revisions
Final Project Plan	10/16/2026	Canvas Submission/GitHub Repository	Deadline given 2 days before actual Canvas deadline to give time for editing and revisions
AI Usage and Team Report	10/16/2026	Canvas Submission/GitHub Repository	Deadline given 2 days before actual Canvas deadline to give time for editing and revisions

1.3. Assumptions, Dependencies, and Constraints

ID	Description
AS-1	The Preliminary Definition, templates, and samples posted on Canvas remain during Phase I.

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ID	Description
AS-2	Each team member can commit approximately 2-3 hours per week to the project, including team meetings.
AS-3	The team may not be able to interview actual BVI users. If not, the team will use personas and published accessibility guidance to represent users' needs.
AS-4	A mockup prototype composed of UI sketches is sufficient for Phase I.
DE-1	Due dates for all deliverables are set by the Course Schedule.
DE-2	The final Project Plan depends on instructor feedback on the preliminary submission.
DE-3	The AS-IS/TO-BE scenarios depend on the Lesson 6 material and the Theia-AS-IS_and_TO-BE_v2.0.pptx template; the WRS depends on the WRS template, incomplete sample, and complete example document.
DE-4	The team depends on third-party services (GitHub, YouTube, recording software, Gen AI tools) being available.
CO-1	Schedule: deadlines are fixed by the course and cannot be moved by the team.
CO-2	Staff: the team size is fixed at 5 members for the semester.
CO-3	Cost: there is no budget; only free or university-licensed tools will be used.
CO-4	The recorded presentation must be no more than 10 to 15 minutes long.
CO-5	Only Phase I deliverables are in scope; hypothetical deliverables such as test plans or design documents are excluded.
CO-6	Any Gen AI use must be disclosed with evidence, and the team remains fully responsible for the correctness of all submissions.

Five Project Dimensions Table.

Dimension	Classification	Rationale
Schedule	Constraint	Due dates are fixed by the Course Schedule (CO-1).
Staff	Constraint	Team membership is fixed (CO-2).
Cost	Constraint	No budget is available (CO-3).
Quality	Success driver	Deliverables are graded on completeness, consistency, traceability, and coherence.
Features	Degree of freedom	The team decides which missing requirements to add and how to resolve each issue, guided by its design rationale.

1.4. References

- Cpt S 484 Team Project Specification, Phase I: Requirements Elicitation, Initial Understanding, v2.0 (adopted from Dr. Lawrence Chung's course at UTD). Canvas.
- Cpt S 484 Project THEIA: Preliminary Definition. Canvas.

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- WRS document template, incomplete sample, and complete example document for another project. Canvas.
- Theia-AS-IS_and_TO-BE_v2.0.pptx presentation template. Canvas.
- Cpt S 484 Lesson 6: AS-IS/TO-BE scenarios. Canvas.
- Cpt S 484 Course Schedule. Canvas.
- Project Management Plan template (ProjectPlanTemplate_updated.docx) and sample project plans. Canvas.
- F. P. Brooks, "No Silver Bullet: Essence and Accidents of Software Engineering," 1987.
- Team Git Repository:

1.5. Definitions and Acronyms

Term / Acronym	Definition
THEIA	The generic name of the app (after the Greek goddess of vision): a smartphone app that helps blind and visually impaired people navigate indoors.
BVI	Blind and Visually Impaired: the primary users of THEIA.
WRS	World, Requirements, Specification. The main Phase I document, which describes the domain (World), the requirements, and the system specification, and the traceability between them.
AS-IS scenario	A description of how a task is performed today, without the proposed system, including its problems.
TO-BE scenario	A description of how the same task will be performed with the proposed system.
FR / NFR	Functional Requirement / Non-Functional Requirement.
RS	Requirement Specification: the most detailed level of the WRS hierarchy, describing specific features of the app.
Traceability	The documented links between Problems, Goals, Objectives, and Requirement Specifications.
Mock-up	A low-fidelity prototype (drawings or a GUI demonstration) showing how users will interact with the app.
Gen AI	Generative artificial intelligence tools, such as large language model chat assistants.

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2. Project Organization

The team's main external entities are the course instructor and the TAs, who act as the evaluator for each deliverable; the Team Liaison will handle communication with them. Deliverables will be submitted through Canvas and GitHub.

2.1. Process Model

The team will follow an iterative requirement engineering process. The phases below overlap, with some phases requiring parts from previous phases to build off and polish.

Phase	Key Tasks	Outputs
1. Planning and team setup	Set up git repository and communication channel; read the specification, Preliminary Definition, and templates; assign tasks.	Preliminary Project Plan
2. Elicitation and analysis	Analyze each statement of the Preliminary Definition; classify issues; propose resolutions with design rationale.	WRS: Issues
3. Specification	Identify Problems; derive Goals, functional and non-functional Objectives, and Requirement Specifications; build the traceability matrix; clarify World and System descriptions.	WRS: Improved Understanding
4. Validation	Write AS-IS/TO-BE scenario pairs and Top Priority analysis; build the mock-up; write the user manual; peer-review all documents for consistency and traceability.	Slides, prototype, user manual
5. Presentation and submission	Record and upload the presentation; compile the AI Usage and Teamwork Report; finalize the Project Plan; submit.	Video link, AI report, final Project Plan

2.2. Organizational Structure

The team is a small, self-organizing group of 5 members. The Team Liaison will ultimately be in charge of coordinating the work to complete the objectives the day before they are due; every other member is responsible for splitting up the work evenly and completing the objectives. Each deliverable will have one owner and one reviewer.

- **Meetings:** team meetings on a need basis. Minutes, decisions, and future actions are recorded in the GitHub Repository
- **Communication:** communication will occur through outlook. Members are expected to reply within 24 hours.
- **Work tracking:** tasks are tracked on GitHub, as well as milestones to be communicated through team through outlook
- **Decision making:** decisions are made through total consensus.

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2.3. Roles and Responsibilities

Role	Member	Responsibilities
Team Liaison / Project Manager	Jacob Atwood	Single point of contact with the instructor; schedules and runs meetings; maintains this Project Plan and the task board; tracks risks; performs the cross-section consistency review of the Issues; submits deliverables on Canvas.
Requirements Analyst (Issues Lead)	Jacob Atwood	Leads the Issues part of the WRS; merges issues from all members into one consistent format; checks that every statement in the Preliminary Definition has been analyzed.
Requirements Analyst (Specification and Traceability Lead)	Eniola	Leads the Improved Understanding; owns the Problem, Goal, Objective, and RS identifiers and the traceability matrix.
UX and Accessibility Designer	David Acosta Leon	Leads the mock-up prototype and the preliminary user manual; ensures the design supports non-visual interaction.
Presentation Lead	Jaydon Devictoria	Leads the AS-IS/TO-BE slides and Top Priority analysis; coordinates recording, editing, timing (10 to 15 minutes)
Configuration Manager and Document Editor	Travis Takushi	Maintains the git repository structure and naming conventions; performs the final editing pass so that all documents read as one coherent submission.
User Proxy / Subject Matter Expert	All members	Contact potential users where possible; otherwise role-play BVI users and review requirements from their point of view.

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3. Managerial Process Plans

This section describes how the team will manage its work, risks, and schedule during Phase I.

3.1. Management objectives and priorities

- **Meet every deadline.** Each deliverable will have a due date of 1 day before the Canvas due date, reserved for review
- **Maximize quality of WRS document.** Document must read as one team's work, as well as holds the most weight of the grade
- **Keep workload balanced and visible.** Work is assigned through discussion, and will be reviewed after every milestone to rebalance

3.2. Assumptions, dependencies, and constraints

- Members are expected to respond to direct outlook messages within 24 hours
- Individual contributions are recorded through git commits, the task board, and meeting minutes; these records are also used for the Teamwork part of the AI Usage and Teamwork Report.
- If a member is unresponsive for more than 3 days, the Team Liaison reassigns their tasks; if the problem continues, the Liaison informs the instructor.
- All deliverables are stored in the team git repository before they are submitted on Canvas.

3.3. Risk management

ID	Risk	Prob.	Impact	Mitigation / Contingency
R-1	A member becomes unavailable (illness, workload, dropping the course).	M	H	Every deliverable has a reviewer who knows its content; the Liaison reassigns tasks within 24 hours.
R-2	Sections written by different members are inconsistent in terminology, identifiers, or style.	H	H	Shared glossary and ID scheme agreed in week 1; final editing pass by the Document Editor.
R-3	Gen AI output contains errors or hallucinations that reach a submission.	M	H	Every AI-assisted passage is checked against the specification and Preliminary Definition by a second member; AI use is logged.
R-4	The team misinterprets the scope or format of a deliverable.	M	H	Use the preliminary plan feedback; compare drafts with the samples; ask the instructor early through the Liaison.

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ID	Risk	Prob.	Impact	Mitigation / Contingency
R-5	Traceability gaps (requirements with no objective, or objectives with no requirement).	M	H	Maintain the traceability matrix as the WRS is written, not at the end; check it before each review.
R-6	Git merge conflicts or lost work on binary files (Word, PowerPoint).	M	M	One editor per file at a time; commit often; keep dated backups of binary files.
R-7	Scope creep (adding out-of-scope deliverables or too many features).	L	M	Check every new item against the specification and constraint CO-5.

3.4. Schedule and Milestones

Wk	Dates	Activities	Owner(s)	Milestone
1	9/20/2026	Set up git repository; read the specification, Preliminary Definition, and templates; agree on glossary and ID scheme; draft the Project Plan.	all	Preliminary Project Plan submitted
2	10/12/2026	Analyze each section of the Preliminary Definition; attempt to contact potential users; create personas; start the issue log.	Travis Takushi Eniola: preliminary definition	Issue log started
3	10/12/2026	Classify and resolve issues with design rationale; cross-review sections; incorporate instructor feedback on the plan.	all	Issues draft complete
4	10/16/2026	Write Problems, Goals, Objectives, and Requirement Specifications; build the traceability matrix; draft four AS-IS/TO-BE pairs.	Jaydon: scenario pairs only	Improved Understanding draft
5	10/16/2026	Build the mock-up prototype; write the user manual from the TO-BE scenarios; prepare slides and Top Priority analysis.	David: prototype and manual; Jaydon: slides	None
6	10/16/2026	Final Draft of WRS	Eniola: Issues, Jacob: Usability	Full WRS draft

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Wk	Dates	Activities	Owner(s)	Milestone
6	10/17/2026	Full-team review and editing pass; rehearse and record the presentation; compile the AI Usage and Teamwork Report; finalize the Project Plan.	All	Slides final draft, Full WRS final draft, AI and Teamwork final draft, final recording

4. Technical Process Plans

Phase I is for requirements engineering, focused on analyzing and validating requirements rather than writing code. The only technical implementation will be the mock-up prototype.

4.1. Methods, tools, and techniques

- **Software Tools:** GitHub: version control and tasks, Microsoft Word: drafting, PowerPoint: presentations for AS-IS/TO-BE, Zoom: recording/editing, Canvas: submitting project deliverables
- **Requirements methods:** document analysis of Preliminary Definition, team brainstorming for models, Issues are classified as incompleteness, inconsistency, ambiguity, redundancy, or unsoundness, and each resolution records the chosen option, the alternatives considered, and the reason for the choice
- **Specification conventions:** the WRS template structure (World, Requirements, Specification); a Problem, Goal, Objective, and Requirement Specification hierarchy with unique IDs (for example P-1, G-1, FO-1, NFO-1, FR-1, NFR-1); a traceability matrix linking all levels; AS-IS/TO-BE scenarios following Lesson 6; and a shared glossary for consistent terminology.
- **Prototype approach:** the mock-up treats audio feedback, haptic (vibration) cues, voice commands, and screen-reader compatibility as the primary interface, with large, high-contrast visual elements as a secondary aid for low-vision users.

4.2. Software documentation

Document	Template or Standard	Created By	Reviewed By	Target Date	Distribution
Project Plan (preliminary and final)	ProjectPlanTemplate_updated.docx; course samples	All(Preliminary); All(Final)	All	9/20/2026	Canvas; git

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Document	Template or Standard	Created By	Reviewed By	Target Date	Distribution
WRS Document (Issues and Improved Understanding)	WRS template, sample, and example on Canvas	Travis Takushi (Issues), Eniola (Improved Understanding)	Travis Takushi	TBA	Canvas; git
AS-IS/TO-BE slides and recorded presentation	Theia-AS-IS_and_TO-BE_v2.0.pptx; Lesson 6	Jaydon Devictoria; all members	All	10/17/2026	Slides in git; video on YouTube (unlisted), link on Canvas
Mock-up prototype	Team conventions; platform accessibility guidelines	David Acosta Leon	Jaydon Devictoria	10/17/2026	Canvas; GitHub
Preliminary user manual	WRS template section (or standalone)	David Acosta Leon	Jaydon Devictoria	10/17/2026	Canvas; GitHub
AI Usage and Teamwork Report	No template; includes evidence	Travis Takushi		TBA	Canvas; GitHub
Meeting minutes (internal)	Team minutes format	Rotating		Request Basis	GitHub only

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